

Duality

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Why should I bother to learn this stuff?

- Duality allows a second representation of the same convex problem, giving sometimes some interesting insights (e.g. principle of virtual forces in mechanics)
- Duality is a good way of obtaining lower bounds
- Duality is a powerful tool for decomposition methods
- $\implies$ fundamental both for studying optimization (continuous and operations research)
- $\implies$ useful in other fields like mechanics and machine learning

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- 1 Lagrangian duality [BV 5]
- 2 Strong duality
- 3 Marginal interpretation of the multiplier
- 4 Revisiting the KKT conditions
- 5 Wrap-up



Consider the following problem

$$\min_{x \in \mathcal{X}} \sup_{y \in \mathcal{Y}} \Phi(x, y)$$

where, for the moment, $\mathcal{X}$ and $\mathcal{Y}$ are arbitrary sets, and Φ an arbitrary function.
By definition the dual of this problem is

$$\max_{y \in \mathcal{Y}} \inf_{x \in \mathcal{X}} \Phi(x, y)$$

and we have **weak duality**, that is

$$\sup_{y \in \mathcal{Y}} \inf_{x \in \mathcal{X}} \Phi(x, y) \leq \inf_{x \in \mathcal{X}} \sup_{y \in \mathcal{Y}} \Phi(x, y)$$

♣ Exercise: Prove this result. Getting equality is more tricky...

Dual representation of some characteristic functions

Recall that, if $X \subset \mathbb{R}^n$

$$\mathbb{I}_X(x) = \begin{cases} 0 & \text{if } x \in X \\ +\infty & \text{otherwise} \end{cases}$$

and if X is an assertion,

$$\mathbb{I}_X = \begin{cases} 0 & \text{if } X \\ +\infty & \text{otherwise} \end{cases}$$

Note that

$$\mathbb{I}_{g(x)=0} = \sup_{\lambda \in \mathbb{R}^{n_E}} \lambda^\top g(x)$$

and

$$\mathbb{I}_{h(x) \leq 0} = \sup_{\mu \in \mathbb{R}_+^{n_I}} \mu^\top h(x)$$

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From constrained to min-sup formulation



$$\begin{array}{ll} \text{Min}_{x \in \mathbb{R}^n} & f(x) \\ \text{s.t.} & g_i(x) = 0 \quad \forall i \in [n_E] \\ & h_j(x) \leq 0 \quad \forall j \in [n_I] \end{array} \quad (P)$$

Is equivalent to

$$\text{Min}_{x \in \mathbb{R}^n} \quad f(x) + \mathbb{I}_{g(x)=0} + \mathbb{I}_{h(x) \leq 0}$$

or

$$\text{Min}_{x \in \mathbb{R}^n} \quad f(x) + \sup_{\lambda \in \mathbb{R}^{n_E}} \lambda^\top g(x) + \sup_{\mu \in \mathbb{R}_+^{n_I}} \mu^\top h(x)$$

which is usually written

$$\text{Min}_{x \in \mathbb{R}^n} \quad \sup_{\lambda, \mu \geq 0} \underbrace{f(x) + \lambda^\top g(x) + \mu^\top h(x)}_{:= \mathcal{L}(x; \lambda, \mu)}$$

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Lagrangian duality



To a (primal) problem (no convexity or regularity assumptions here)

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we associate the **Lagrangian**

$$\mathcal{L}(x; \lambda, \mu) := f(x) + \lambda^\top g(x) + \mu^\top h(x)$$

such that

$$(P) \quad \underset{x \in \mathbb{R}^n}{\text{Min}} \quad \sup_{\lambda, \mu \geq 0} \mathcal{L}(x; \lambda, \mu)$$

The dual problem is defined as

$$(D) \quad \underset{\lambda, \mu \geq 0}{\text{Max}} \quad \inf_{x \in \mathbb{R}^n} \mathcal{L}(x; \lambda, \mu)$$

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Weak duality

By the min-max duality, we easily see that

$$\text{val}(D) \leq \text{val}(P).$$

Further any admissible dual multipliers $\lambda \in \mathbb{R}^{n_E}$ $\mu \in \mathbb{R}_+^{n_I}$ yields a lower bound:

$$g(\lambda, \mu) := \inf_{x \in \mathbb{R}^n} \mathcal{L}(x; \lambda, \mu) \leq \text{val}(D) \leq \text{val}(P)$$

Obviously, any admissible solution $x \in \mathbb{R}^n$ (i.e. such that $g(x) = 0$ and $h(x) \leq 0$), yields an upper bound

$$\text{val}(P) \leq f(x) = \sup_{\lambda, \mu \geq 0} \mathcal{L}(x; \lambda, \mu)$$

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Min-Max duality

Recall the generic primal problem of the form

$$p^* := \underset{x \in \mathcal{X}}{\text{Min}} \quad \sup_{y \in \mathcal{Y}} \quad \Phi(x, y)$$

with associated dual

$$d^* := \underset{y \in \mathcal{Y}}{\text{Max}} \quad \inf_{x \in \mathcal{X}} \quad \Phi(x, y).$$

Recall that the **duality gap** $p^* - d^* \geq 0$.

We say that we have **strong duality** if $d^* = p^*$.

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Saddle point

Definition

Let $\Phi : \mathcal{X} \times \mathcal{Y} \rightarrow \overline{\mathbb{R}}$ be any function. $(x^\sharp, y^\sharp)$ is a (local) saddle point of Φ on $\mathcal{X} \times \mathcal{Y}$ if

- $x^\sharp$ is a (local) minimum of $x \mapsto \Phi(x, y^\sharp)$.
- $y^\sharp$ is a (local) maximum of $y \mapsto \Phi(x^\sharp, y)$.

If there exists a global Saddle Point $(x^\sharp, y^\sharp)$ of Φ , then there is strong duality, $x^\sharp$ is an optimal primal solution and $y^\sharp$ an optimal dual solution, i.e.

$$p^* = d^* = \Phi(x^\sharp, y^\sharp).$$

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Theorem (Von Neumann Minimax Theorem)

If

- $\mathcal{X}$ and $\mathcal{Y}$ are convex, one of them is compact
- Φ is continuous on $\mathcal{X} \times \mathcal{Y}$
- $\Phi(\cdot, y)$ is convex for all $y \in \mathcal{Y}$
- $\Phi(x, \cdot)$ is concave for all $x \in \mathcal{X}$

then there exists a saddle point of Φ

$\leadsto$ we can exchange "Min" and "Max".

Slater's conditions for convex optimization



Consider the following **convex** optimization problem

$$\begin{aligned} (P) \quad & \underset{x \in \mathbb{R}^n}{\text{Min}} && f(x) \\ & \text{s.t.} && Ax = b \\ & && h_j(x) \leq 0 \qquad \forall j \in [n_I] \end{aligned}$$

We say that a point x^s such that $Ax^s = b$, $x^s \in \text{ri}(\text{dom}(f))^1$, and $h_j(x^s) < 0$ for all $j \in [n_I]$, is a **Slater's point**.

¹for extended function f

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Theorem

If (P) is convex (i.e. f and h_j are convex), and there exists a Slater's point then there is strong (Lagrangian) duality.

Further if (P) admits an optimal solution $x^\#$ then $\mathcal{L}$ admits a saddle point $(x^\#, \lambda^\#)$, and $\lambda^\#$ is an optimal solution to (D) .

¹for extended function f

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Perturbed problem



We consider the following perturbed problem

$$\begin{aligned} v(p, q) = \quad & \text{Min}_{x \in \mathbb{R}^n} \quad f(x) \\ & \text{s.t.} \quad g(x) = p \\ & \quad \quad h(x) \leq q \end{aligned}$$

In particular we have $v(0, 0) = \text{val}(P)$.

By duality,

$$v(p, q) \geq d(p, q) = \sup_{\lambda, \mu \geq 0} \inf_x f(x) + \lambda^\top (g(x) - p) + \mu^\top (h(x) - q).$$

In particular, d is convex as a supremum of convex functions.



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Marginal interpretation of the dual multiplier



Assume that (P) is convex, and satisfies the Slater's qualification condition. In particular $v(0, 0) = d(0, 0)$.

Let (λ, μ) be optimal multipliers of (P) .

We have, for any $x_{p,q}$ admissible for the perturbed problem, that is such that $g(x_{p,q}) = p$, $h(x_{p,q}) \leq q$,

$$\begin{aligned}\text{val}(P) = v(0, 0) &= \inf_x f(x) + \lambda^\top g(x) + \mu^\top h(x) \\ &\leq f(x_{p,q}) + \lambda^\top g(x_{p,q}) + \mu^\top h(x_{p,q}) \\ &\leq f(x_{p,q}) + \lambda^\top p + \mu^\top q\end{aligned}$$

In particular we have,

$$v(p, q) = \inf_{x_{p,q}} f(x_{p,q}) \geq v(0, 0) - \lambda^\top p - \mu^\top q$$

which reads

$$-(\lambda, \mu) \in \partial v(0, 0)$$

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Exercise

♣ Exercise: Consider the following problem, for $b \in \mathbb{R}$,

$$\begin{array}{ll} \text{Min} & x^2 \\ \text{s.t.} & x \leq b \end{array}$$

- 1 Does there exist an optimal multiplier?
- 2 Without solving the dual, give the optimal multiplier μ_b .

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Recall the first order KKT conditions for our problem (P)

$$\nabla f(\mathbf{x}) + A^T \lambda + \sum_{j=1}^{n_I} \mu_j \nabla h_j(\mathbf{x}) = 0$$

$$A\mathbf{x} = b, \quad h(\mathbf{x}) \leq 0$$

$$\lambda \in \mathbb{R}^{n_E}, \quad \mu \in \mathbb{R}_+^{n_I}$$

$$\mu_j g_j(\mathbf{x}) = 0$$

$$\forall j \in [n_I]$$

Further, recall that

- the existence of a Slater's point in a convex problem ensures constraints qualifications,
- first order conditions are sufficient for convex problems.



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Further, recall that

- the existence of a Slater's point in a convex problem ensures constraints qualifications,
- first order conditions are sufficient for convex problems.



If (P) is convex and there exists a Slater's point. Then the following assertions are equivalent:

- 1 $x^\#$ is an optimal solution of (P) ,
- 2 $(\exists \lambda^\# \text{ such that}) (x^\#, \lambda^\#)$ is a saddle point of $\mathcal{L}$,
- 3 $(\exists \lambda^\# \text{ such that}) (x^\#, \lambda^\#)$ satisfies the KKT conditions.



$$\begin{aligned}
 (P) \quad & \underset{x \in \mathbb{R}^n}{\text{Min}} \quad f(x) \\
 & \text{s.t.} \quad A(x) = b \\
 & \quad \quad h_j(x) \leq 0 \quad \forall j \in [n_I]
 \end{aligned}$$

with associated Lagrangian

$$\mathcal{L}(x; \lambda, \mu) := f(x) + \lambda^\top (Ax - b) + \mu^\top h(x)$$

The KKT conditions can be seen as:

- ① $\nabla_x \mathcal{L}(x; \lambda, \mu) = 0$ (Lagrangian minimized in x)
- ② $g(x) = 0, h(x) \leq 0$ (x primal admissible, also obtained as $\nabla_\lambda \mathcal{L} = 0$)
- ③ $\mu \geq 0$ ((λ, μ) dual admissible)
- ④ $\mu_j = 0$ or $h_j(x) = 0$, for all $j \in [n_I]$ (complementarity constraint $\leadsto 2^{n_I}$ possibilities).



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Consider a convex problem satisfying Slater's condition.

Recall that $-\mu^\sharp \in \partial v(0)$ where $v(p)$ is the value of the perturbed problem. From this interpretation, we can recover the complementarity condition

$$\mu_j = 0 \quad \text{or} \quad h_j(x) = 0$$

Indeed, let x be an optimal solution.

- If constraint j is not saturated at x (i.e. $h_j(x) < 0$), we can marginally move the constraint without affecting the optimal solution, and thus the optimal value. In particular, it means that $\mu_j = 0$.
- If $\mu_j \neq 0$, it means that marginally moving the constraint changes the optimal value and thus the optimal solution. In particular, constraint j must be saturated, i.e. $h_j(x) = 0$.

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What you have to know

- Weak duality: $\sup \inf \Phi \leq \inf \sup \Phi$
- Definition of the Lagrangian $\mathcal{L}$
- Definition of primal and dual problem

$$\underbrace{\text{Max}_{\lambda, \mu} \inf_x \mathcal{L}(x; \lambda, \mu)}_{\text{Dual}} \leq \underbrace{\text{Min}_x \sup_{\lambda, \mu} \mathcal{L}(x; \lambda, \mu)}_{\text{Primal}}$$

- Marginal interpretation of the optimal multipliers

What you really should know

- A saddle point of $\mathcal{L}$ is a primal-dual optimal pair
- Sufficient condition of strong duality under convexity (Slater's)

What you have to be able to do

- Turn a constrained optimization problem into an unconstrained Min sup problem through the Lagrangian
- Write the dual of a given problem
- Heuristically recover the KKT conditions from the Lagrangian of a problem

What you should be able to do

- Get lower bounds through duality